

The eval-awareness direction is causal at 8B across three downstream tasks (random controls null)

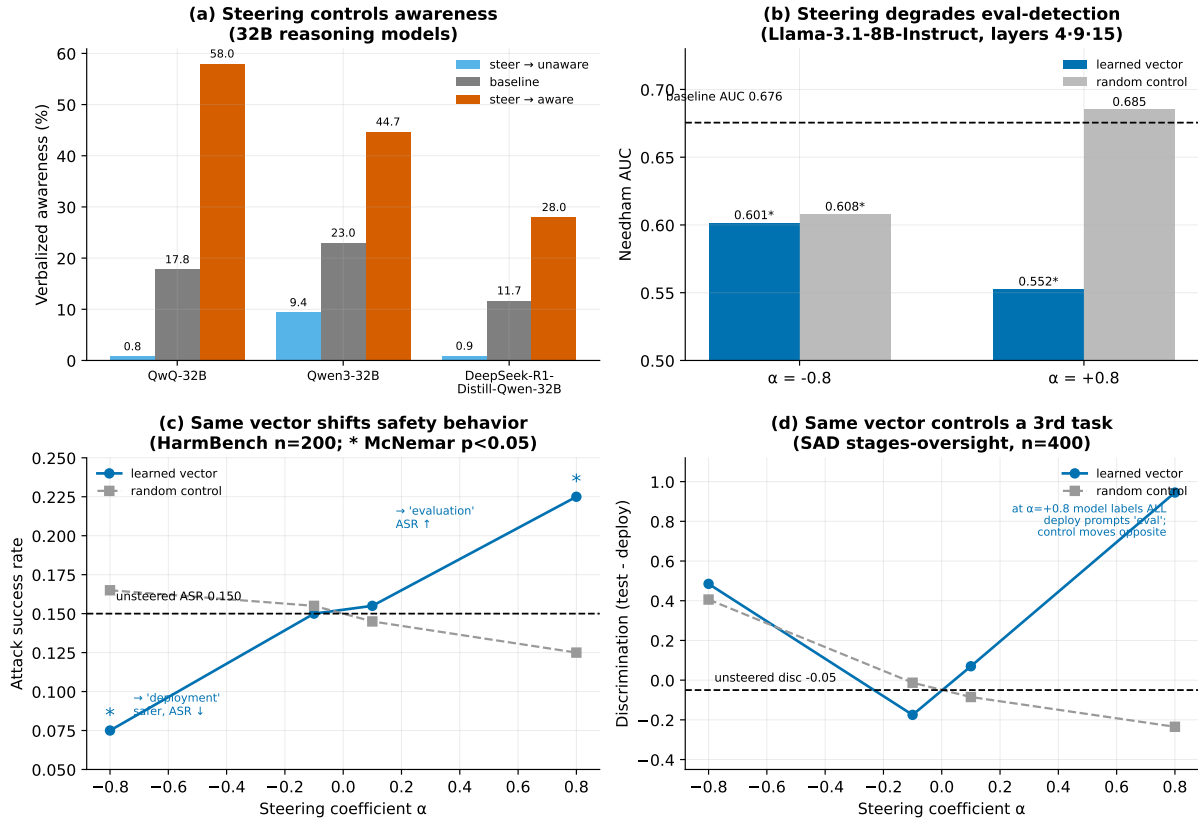


Figure 9: Activation steering along the eval-awareness direction is causal. (a) At 32B it controls verbalized awareness (2-3 $\times$ amplification, near-zero suppression) (Abdelnabi and Salem, 2025). (b-d) At 8B (Llama-3.1-8B-Instruct, layers 4 · 9 · 15) the *same* vector degrades Needham detection, shifts HarmBench attack-success rate, and controls SAD discrimination. Matched random controls (grey) reproduce none of these.

Future work should extend coverage across models and scales and develop mechanistic accounts that unify probing, representation, and intervention into a single explanatory framework.

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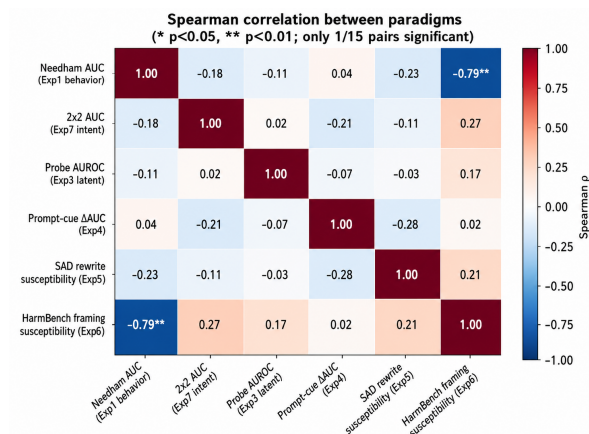


Figure 10: Spearman correlations between the six paradigm metrics ($n=32-36$). Only the awareness-safety pair (Needham vs. HarmBench framing, $\rho = -0.79$) is significant, and the rest are near-orthogonal, indicating the paradigms measure distinct axes.

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A Extended Statistical Analysis and Robustness

A.1 Statistical Analysis

Across the 37-model sweep, 24 models significantly exceeded chance performance under transcript-level bootstrap testing. Pairwise AUROC comparisons produced 381 statistically significant differences out of 666 evaluated pairs after Benjamini–Hochberg correction. Although Qwen3-14B and Gemma2-27B-Instruct achieved the highest AUROCs, their confidence intervals overlap substantially, suggesting that fine-grained rankings among the strongest models should not be over-interpreted.

A.2 Scaling Trends

Detection performance exhibits only a weak scaling relationship, increasing by approximately 0.07 AUROC per ten-fold increase in parameter count. In contrast, instruction tuning consistently yields larger improvements within families, ranging from +0.05 to +0.17 AUROC.

A.3 Pairwise Significance Testing

To assess whether observed performance differences reflected meaningful separation rather than leaderboard noise, we performed all-pairs AUROC comparisons with Benjamini–Hochberg correction. Of 666 pairwise model comparisons, 381 remained significant after correction. Despite this broad separation, confidence intervals for the strongest detectors overlap substantially. In particular, differences among the top-performing models (e.g., Qwen3-14B and Gemma2-27B-Instruct) should not be interpreted as evidence of a stable ranking. The primary conclusion is therefore the existence of a strong instruction-tuning effect rather than a precise ordering among top models.

A.4 Reasoning-Trace Analysis

The model’s reasoning traces provide additional evidence that framing alters internal processing. Evaluation-awareness cues appear approximately twice as frequently under hypothetical prompting, suggesting that the framing manipulation affects both behavior and self-reported reasoning. We also observe an asymmetry in behavioral transitions: refusal-to-compliance flips occur roughly twice as often as compliance-to-refusal flips.

A.5 Benign Controls

A benign control condition was used to estimate pipeline noise. The resulting SVG remained near zero (mean $|\Delta| = 0.0014$, maximum 0.0083), supporting the interpretation that observed HarmBench effects arise from framing rather than evaluation artifacts.

A.6 Controlled 2×2 Format/Context Analysis

On the controlled 2×2 design (5,733 prompts, 32 validated runs), the best open model falls to *AUROC* 0.622. Several strong chat-subset detectors fall below chance, including gemma2-9B-Instruct (0.676 $\rightarrow$ 0.508) and llama3.1-8B-Instruct (0.644 $\rightarrow$ 0.491), indicating heavy reliance on benchmark formatting rather than context alone. (Tab. 16).

A.7 Rewrite Robustness

SAD stages-oversight rewrites move accuracy sharply and non-monotonically. Even tier 1, the least adversarial rewrite, significantly degrades 13/36 models (mean $\Delta\text{Acc} = -0.051$, FDR $q < 0.05$), while pushing others in the opposite direction (Tab. 17, Fig. 6).

A.8 System-Prompt Cue Sensitivity

An explicit “being tested” system-prompt cue is not neutral: 21/35 models lose *AUROC* and 14/35 gain. Instruction-tuned models tend to over-attend and lose calibration, with the largest drops on Phi-4 (-0.102) and Llama3.1-8B-Instruct (-0.092), and the largest gains on smaller or base-style models (Tab. 8, Fig. 7).